

HCSM-10

The Mirror Cancellation Theorem

Residual 7/32 and the Emergent Standard Model Generation from a Single Lattice Primitive

The Hodge Complex Standard Model

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Abstract

We prove that a single primitive the existence of a finite periodic square lattice forces the entire Hodge Complex Standard Model (HCSM) substrate, including the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$, the 14-mode multiplet at the self-paired eigenvalue $\lambda = 4$, the exact residual ratio 7/32, and the emergent Standard Model generation. The mirror symmetry $\lambda \leftrightarrow 8 - \lambda$ of the 5-point Laplacian spectrum pairs 50 of the 64 modes into cancelling partners; only the 14 self-paired modes at $\lambda = 4$ survive. For the zero-point convention $F(\lambda) = \lambda/2$, the residual contribution is 28 out of 128, giving the ratio 7/32. We further prove three operator algebraic theorems: the composite involution $TF = T_{\text{half}} \circ F$ grades the 14-mode multiplet by $|\Delta n|$; the TF-fixed status is constant on each T_{half} pair and coincides with the HCSM-23 within-pair ordering rule; and the four mode set $\{u, s, \text{dark}, \tau\}$ carries a faithful Klein four group action generated by F and TF . All results are derived from the single lattice primitive via the entropy-spacing selection of $L = 8$. The physical reading of 7/32—the identification of the surviving 14 modes with the residual dark-energy density $\rho_{\text{DE}} = (d+1)/2 \cdot m_\nu^4$, matching observation to 0.010%- is derived in the companion paper HCSM-53. Every numerical claim is verified in IEEE-754 double precision and is exact in rational arithmetic.

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1 Introduction

The Hodge Complex Standard Model (HCSM) derives the gauge structure, matter content, particle spectrum, and mixing matrices of the Standard Model from a single discrete substrate: the periodic 8×8 torus $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ [1, 2]. The central physical claim is that the Standard Model is the residue left when a small, discrete, information-bearing lattice cancels against its own mirror.

This paper establishes the mathematical foundation of that cancellation. We prove that a *single* primitive forces the entire substrate, the 14-mode multiplet, the residual ratio $7/32$, and the emergent Standard Model generation. The physical reading of the residual ratio as the observed dark-energy density is the subject of the companion paper HCSM-53 [12].

1.1 The single primitive

The paper rests on exactly one axiom.

Axiom 1 (Lattice primitive). There exists a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L a positive even integer.

Every object in the paper the substrate size $L = 8$, the 5-point Laplacian, the Hodge complex, the mirror symmetry, the 14-mode multiplet, and the residual $7/32$ - is a theorem of ?? 1. No physical input, no empirical anchor, and no fitted parameter is used.

1.2 Summary of results

The main theorems are:

- (1) The 5-point Laplacian is the unique D_4 -invariant, nearest neighbour, positive semidefinite operator on Λ_L with kernel equal to the constants (Theorem 2.1).
- (2) The Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$ (Theorem 3.4).
- (3) The unique self paired eigenvalue is $\lambda = 4$, with multiplicity $2(L - 1)$ (Theorem 4.2).
- (4) At $L = 8$, the 64 modes decompose as $50 + 14$ (Corollary 5.2).
- (5) For any function F satisfying $F(\lambda) + F(8 - \lambda) = c$, the paired contribution is $25c$ and the residual contribution is $14F(4)$ (Theorem 6.1).
- (6) For the zero point convention $F(\lambda) = \lambda/2$, the residual ratio is exactly $7/32$ (Theorem 7.1).
- (7) The ratio $7/32$ is convention-independent for any affine $F(\lambda) = \alpha\lambda + \beta$ with $F(4) \neq 0$ (Theorem 8.1).
- (8) The composite involution $\text{TF} = T_{\text{half}} \circ F$ grades the 14-mode multiplet by $|\Delta n|$ (Theorem 9.4).
- (9) TF-fixed status is constant on each T_{half} pair and coincides with the HCSM-23 within-pair ordering rule (Corollary 9.5).
- (10) The four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful Klein four group action generated by F and TF (Theorem 10.2).

1.3 Organization

Section 2 establishes the substrate and the 5-point Laplacian from ?? 1. Section 3 proves the mirror symmetry. Section 4 proves uniqueness and multiplicity of $\lambda = 4$. Section 5 establishes the $50 + 14$ decomposition. Section 6 proves the cancellation theorem. Section 7 derives $7/32$. Section 8 proves convention independence. Section 9 establishes the TF operator and the $|\Delta n|$ grading. Section 10 proves the Klein four orbit structure. Section 11 derives $L = 8$ from ?? 1 via entropy spacing. Section 12 states the physical reading via HCSM-53. Section 13 catalogs inputs and derived results. Section 14 gives reproducibility notes. Section 15 concludes.

2 Setup

2.1 The lattice

By ?? 1, the substrate is $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L even. Site (x_1, x_2) is identified with $(x_1 + L, x_2)$ and with $(x_1, x_2 + L)$. The lattice carries the D_4 point group of the square, generated by

$$r : (n_1, n_2) \mapsto (-n_2, n_1) \pmod{L}, \quad s : (n_1, n_2) \mapsto (n_1, -n_2) \pmod{L}, \quad (1)$$

with presentation $r^4 = s^2 = (rs)^2 = e$.

2.2 The 5-point Laplacian

Theorem 2.1 (Laplacian uniqueness). *Let Δ be a linear operator on Λ_L satisfying:*

- (i) *D_4 -invariance: $[R, \Delta] = 0$ for every $R \in D_4$;*
- (ii) *nearest neighbor locality: $(\Delta f)(x)$ depends only on $f(x)$ and f at the four nearest neighbors of x ;*
- (iii) *positivity and kernel: Δ is positive-semidefinite and $\ker \Delta$ is exactly the set of constant functions.*

Then Δ is uniquely determined up to an overall positive scale, and equals the 5-point Laplacian

$$(\Delta f)(x_1, x_2) = 4f(x_1, x_2) - f(x_1 + 1, x_2) - f(x_1 - 1, x_2) - f(x_1, x_2 + 1) - f(x_1, x_2 - 1). \quad (2)$$

Proof. Condition (ii) writes $\Delta f(x) = a_0 f(x) + \sum_{\mu} a_{\mu} f(x + \mu)$ over the four nearest neighbor vectors $\mu \in \{\pm e_1, \pm e_2\}$. These four directions form a single D_4 orbit, so condition (i) forces $a_{\mu} = a$ for all four. Condition (iii) forces $\Delta \mathbf{1} = 0$, hence $a_0 + 4a = 0$ or $a_0 = -4a$. Normalizing $a = 1$ gives Eq. (2). $\square$

2.3 The Laplacian spectrum

For $(n_1, n_2) \in \{0, \dots, L-1\}^2$, define the plane wave

$$\phi_{n_1, n_2}(x_1, x_2) = \exp\left(\frac{2\pi i(n_1 x_1 + n_2 x_2)}{L}\right). \quad (3)$$

Direct application of Eq. (2) gives the eigenvalue

$$\lambda(n_1, n_2) = 4 - 2 \cos\left(\frac{2\pi n_1}{L}\right) - 2 \cos\left(\frac{2\pi n_2}{L}\right). \quad (4)$$

Corollary 2.2. $\lambda(n_1, n_2) \in [0, 8]$, with minimum 0 attained only at $(0, 0)$ and maximum 8 attained only at $(L/2, L/2)$.

3 Mirror symmetry

Definition 3.1 (Half period translation). The half-period translation is the map

$$T_{\text{half}}(n_1, n_2) = ((n_1 + L/2) \bmod L, (n_2 + L/2) \bmod L). \quad (5)$$

Lemma 3.2. T_{half} is an involution on $\{0, \dots, L-1\}^2$.

Proof. $(n + L/2 + L/2) \bmod L = n \bmod L$ for both coordinates. $\square$

Lemma 3.3 (Cosine shift). For $n \in \{0, \dots, L-1\}$ with L even,

$$\cos\left(\frac{2\pi \cdot ((n + L/2) \bmod L)}{L}\right) = -\cos\left(\frac{2\pi n}{L}\right). \quad (6)$$

Proof. Let $m = (n + L/2) \bmod L$. If $n < L/2$, then $m = n + L/2$ and $\cos(2\pi m/L) = \cos(2\pi n/L + \pi) = -\cos(2\pi n/L)$. If $n \geq L/2$, then $m = n - L/2$ and $\cos(2\pi m/L) = \cos(2\pi n/L - \pi) = -\cos(2\pi n/L)$. Both cases coincide. $\square$

Theorem 3.4 (Mirror symmetry). *For every momentum pair (n_1, n_2) ,*

$$\lambda(T_{\text{half}}(n_1, n_2)) = 8 - \lambda(n_1, n_2). \quad (7)$$

Consequently, the spectrum of Δ on Λ_L is exactly mirror-symmetric: if λ is an eigenvalue with multiplicity m , then $8 - \lambda$ is also an eigenvalue with multiplicity m .

Proof. By Eq. (4), $\lambda(n_1, n_2) = 4 - 2\cos(2\pi n_1/L) - 2\cos(2\pi n_2/L)$. Applying T_{half} to both coordinates and using Lemma 3.3:

$$\begin{aligned} \lambda(T_{\text{half}}(n_1, n_2)) &= 4 - 2(-\cos(2\pi n_1/L)) - 2(-\cos(2\pi n_2/L)) \\ &= 4 + 2\cos(2\pi n_1/L) + 2\cos(2\pi n_2/L) \\ &= 8 - \lambda(n_1, n_2). \end{aligned} \quad (8)$$

T_{half} is a bijection by Lemma 3.2, so it pairs eigenvalues λ and $8 - \lambda$ at equal multiplicity. $\square$

Corollary 3.5. *Every eigenvalue $\lambda \neq 4$ appears in a mirror pair $(\lambda, 8 - \lambda)$ of equal multiplicity.*

4 The self-paired eigenvalue

Definition 4.1. An eigenvalue λ is *self-paired* if $\lambda = 8 - \lambda$.

Theorem 4.2. *The unique self-paired eigenvalue is $\lambda = 4$.*

Proof. $\lambda = 8 - \lambda$ implies $2\lambda = 8$, hence $\lambda = 4$. $\square$

Theorem 4.3 (Multiplicity at $\lambda = 4$). *For every even L , the multiplicity of $\lambda = 4$ is*

$$\text{mult}_{\lambda=4}(\Lambda_L) = 2(L - 1). \quad (9)$$

Proof. The $\lambda = 4$ eigenspace is the set of (n_1, n_2) satisfying

$$\cos\left(\frac{2\pi n_1}{L}\right) + \cos\left(\frac{2\pi n_2}{L}\right) = 0. \quad (10)$$

Using the sum-to-product identity $\cos a + \cos b = 2\cos((a+b)/2)\cos((a-b)/2)$, this becomes

$$2\cos\left(\frac{\pi(n_1 + n_2)}{L}\right)\cos\left(\frac{\pi(n_1 - n_2)}{L}\right) = 0. \quad (11)$$

So either $n_1 + n_2 \equiv L/2 \pmod{L}$ or $n_1 - n_2 \equiv L/2 \pmod{L}$. Each congruence has L solutions in $\mathbb{Z}_L \times \mathbb{Z}_L$. Their intersection is the set of (n_1, n_2) satisfying both, which gives $n_1 = n_2 + L/2$ and $n_1 + n_2 = L/2$ simultaneously, i.e., $2n_1 = L$ and $2n_2 = 0$ modulo L . The intersection has exactly two points: $(0, L/2)$ and $(L/2, 0)$. By inclusion-exclusion,

$$\text{mult}_{\lambda=4} = L + L - 2 = 2(L - 1). \quad (12)$$

$\square$

Remark 4.4. At $L = 8$, Theorem 4.3 gives $\text{mult}_{\lambda=4} = 14$. This is the central 14-mode multiplet of HCSM [5].

5 The 50 + 14 decomposition

Corollary 5.1 (General L). *The L^2 modes of Λ_L decompose as*

$$(L^2 - 2(L - 1)) \text{ paired modes} + 2(L - 1) \text{ residual modes at } \lambda = 4. \quad (13)$$

Proof. Immediate from Corollary 3.5 and Theorem 4.3. $\square$

Corollary 5.2 ($L = 8$). *At $L = 8$,*

$$50 \text{ paired modes} + 14 \text{ residual modes} = 64. \quad (14)$$

6 The cancellation theorem

Theorem 6.1 (General cancellation). *Let $F : \mathbb{R} \rightarrow \mathbb{R}$ satisfy*

$$F(\lambda) + F(8 - \lambda) = c \quad (15)$$

for all λ and some constant $c \in \mathbb{R}$. Then

$$\sum_m F(\lambda_m) = c \cdot \frac{L^2 - 2(L - 1)}{2} + F(4) \cdot 2(L - 1). \quad (16)$$

Proof. Group the L^2 modes into $(L^2 - 2(L - 1))$ paired modes and $2(L - 1)$ residual modes at $\lambda = 4$. Each pair $(\lambda, 8 - \lambda)$ with multiplicity m_p contributes $m_p \cdot c$ by hypothesis. Summing over all pairs gives $c \cdot (L^2 - 2(L - 1))/2$. Adding the residual contribution $2(L - 1) \cdot F(4)$ gives the result. $\square$

Corollary 6.2 ($L = 8$). *At $L = 8$,*

$$\sum_m F(\lambda_m) = 25c + 14F(4). \quad (17)$$

Proof. Substitute $L = 8$: $(64 - 14)/2 = 25$, $2 \cdot 7 = 14$. $\square$

7 The residual ratio 7/32

Theorem 7.1. *For the zero-point convention $F(\lambda) = \lambda/2$ and $L = 8$:*

$$\sum_m F(\lambda_m) = 128, \quad \sum_{m \in \mathcal{R}} F(\lambda_m) = 28, \quad \frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}, \quad (18)$$

where $\mathcal{R}$ is the residual set of 14 modes at $\lambda = 4$.

Proof. Compute $\sum_m \lambda_m$ exactly. Using the trigonometric identity $\sum_{n=0}^7 \cos(\pi n/4) = 0$, we have

$$\begin{aligned} \sum_m \lambda_m &= \sum_{n_1, n_2=0}^7 \left[4 - 2 \cos\left(\frac{\pi n_1}{4}\right) - 2 \cos\left(\frac{\pi n_2}{4}\right) \right] \\ &= 4 \cdot 64 - 2 \cdot 8 \sum_{n=0}^7 \cos\left(\frac{\pi n}{4}\right) - 2 \cdot 8 \sum_{n=0}^7 \cos\left(\frac{\pi n}{4}\right) \\ &= 256 - 0 - 0 = 256. \end{aligned} \quad (19)$$

Therefore $\sum_m \lambda_m/2 = 128$.

The residual contribution is $\sum_{m \in \mathcal{R}} \lambda_m/2 = 14 \cdot (4/2) = 28$.

Ratio: $28/128 = 7/32$. $\square$

Corollary 7.2 (Explicit split). $\sum_{\text{paired}} F(\lambda_m) = 100$ and $\sum_{\mathcal{R}} F(\lambda_m) = 28$. The paired modes contribute 25/32 of the total; the residual contributes 7/32.

8 Convention independence

Theorem 8.1. Let $F(\lambda) = \alpha\lambda + \beta$ with $\alpha > 0$ and $F(4) \neq 0$. Then

$$\frac{\sum_{m \in \mathcal{R}} F(\lambda_m)}{\sum_m F(\lambda_m)} = \frac{7}{32}. \quad (20)$$

If $F(4) = 0$ (equivalently $\beta = -4\alpha$), both numerator and denominator vanish, and the ratio is undefined.

Proof. Compute:

$$\sum_{m \in \mathcal{R}} F(\lambda_m) = 14F(4) = 14(4\alpha + \beta), \quad (21)$$

$$\sum_m F(\lambda_m) = \alpha \sum_m \lambda_m + 64\beta = 256\alpha + 64\beta = 64(4\alpha + \beta). \quad (22)$$

Ratio = $14(4\alpha + \beta)/[64(4\alpha + \beta)] = 14/64 = 7/32$, provided $4\alpha + \beta \neq 0$. If $4\alpha + \beta = 0$, then $F(4) = 0$ and both sums vanish; the ratio is undefined. $\square$

Remark 8.2 (Correction note). An earlier revision claimed the ratio 7/32 holds for $\beta = -4\alpha$. That is exactly the degenerate case. The correct statement is the opposite: the ratio holds for any affine F except the degenerate case $\beta = -4\alpha$.

Corollary 8.3 (Geometric content). The convention-independent content of the cancellation is

$$\frac{\sum_{m \in \mathcal{R}} \lambda_m}{\sum_m \lambda_m} = \frac{56}{256} = \frac{7}{32}. \quad (23)$$

The numerator $14 \cdot 4 = 56$ is multiplicity times the midpoint eigenvalue; the denominator $64 \cdot 4 = 256$ is total mode count times the midpoint eigenvalue.

9 The TF operator and the $|\Delta n|$ grading

Definition 9.1 (Diagonal flip). The diagonal flip is $F : (n_1, n_2) \mapsto (n_2, n_1)$ on Λ_L .

Definition 9.2 (TF operator). The TF operator is the composition

$$\text{TF} := T_{\text{half}} \circ F, \quad \text{TF}(n_1, n_2) = ((n_2 + L/2) \bmod L, (n_1 + L/2) \bmod L). \quad (24)$$

Lemma 9.3. $\text{TF}^2 = \text{id}$.

Proof. $\text{TF}^2(n_1, n_2) = \text{TF}(n_2 + L/2, n_1 + L/2) = (n_1 + L, n_2 + L) = (n_1, n_2) \bmod L$. $\square$

Theorem 9.4 (TF grading by $|\Delta n|$). Let $\Delta n := n_2 - n_1$ on Λ_8 . On the 14 modes at $\lambda = 4$:

- **TF-fixed modes:** all modes with $\Delta n \equiv 4 \pmod{8}$. There are 8 such modes: $\{\nu, e, d, b, \mu, c, Z, H\}$.
- **TF 2-cycle modes:** all modes with $\Delta n \not\equiv 4 \pmod{8}$. There are 6 such modes, forming three 2-cycles: $\{u, \tau\}$, $\{s, \text{dark}\}$, $\{W, t\}$.

Proof. TF-fixed iff $(n_2 + 4) \bmod 8 = n_1$ and $(n_1 + 4) \bmod 8 = n_2$, equivalent to $n_2 - n_1 \equiv 4 \pmod{8}$. By the parity constraint $n_1 \equiv n_2 \pmod{2}$ (which holds on every mode at $\lambda = 4$), the condition $\Delta n \equiv 4 \pmod{8}$ is equivalent to $|\Delta n| = 4$. Direct enumeration of the 14 modes confirms:

- 8 modes with $|\Delta n| = 4$: $\{\nu, e, d, b, \mu, c, Z, H\}$;
- 4 modes with $|\Delta n| = 2$: $\{u, s, \text{dark}, \tau\}$;
- 2 modes with $|\Delta n| = 0$: $\{W, t\}$.

Direct check of the 2-cycles confirms: $\text{TF}(u) = \tau$, $\text{TF}(s) = \text{dark}$, $\text{TF}(W) = t$. $\square$

Corollary 9.5 (TF-fixed $\Leftrightarrow$ higher $N/4$). *On each T_{half} pair, the TF-fixed member is exactly the member with smaller n_1 , which by the HCSM-23 within-pair ordering rule receives the higher $N/4$ label.*

Proof. Let $\{m, m'\}$ be a T_{half} pair with $n_1(m) < n_1(m')$. Then $\text{TF}(m) = T_{\text{half}}(F(m)) = T_{\text{half}}(m') = m$ by direct computation (since T_{half} is the involution pairing m with m' and $F(m) = m'$ for modes in the same T_{half} pair). Hence the smaller- n_1 member is TF-fixed. Since HCSM-23 assigns the higher $N/4$ to the smaller- n_1 member, TF-fixed $\Leftrightarrow$ higher $N/4$. $\square$

Remark 9.6. Corollary 9.5 *derives* the HCSM-23 within-pair ordering rule from the TF operator of the substrate. What was stated as a rule in HCSM-23 is now a theorem of ?? 1.

10 Klein four-orbit structure

Definition 10.1 (Substrate involutions). On Λ_8 , three commuting involutions act on the mode labels:

$$F : (n_1, n_2) \mapsto (n_2, n_1), \quad (25)$$

$$\text{TF} : (n_1, n_2) \mapsto ((n_2 + 4) \bmod 8, (n_1 + 4) \bmod 8), \quad (26)$$

$$F \cdot \text{TF} : (n_1, n_2) \mapsto ((n_1 + 4) \bmod 8, (n_2 + 4) \bmod 8). \quad (27)$$

All three commute pairwise.

Theorem 10.2 (Klein four-orbit). *The four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful action of the Klein four-group $V_4 = \{\text{id}, F, \text{TF}, F \cdot \text{TF}\}$:*

$$F : u \leftrightarrow s, \quad \text{dark} \leftrightarrow \tau, \quad (28)$$

$$\text{TF} : u \leftrightarrow \tau, \quad s \leftrightarrow \text{dark}, \quad (29)$$

$$F \cdot \text{TF} : u \leftrightarrow \text{dark}, \quad s \leftrightarrow \tau. \quad (30)$$

The action is faithful, transitive, and free (no stabilizer).

Proof. Direct computation on the four modes. All three involutions act as products of two disjoint transpositions, pairwise commute, and generate the Klein four-group. No mode is fixed by any non-identity element, so the action is free. $\square$

Corollary 10.3. *The Klein four-orbit $\{u, s, \text{dark}, \tau\}$ is generated by the substrate involutions F and TF . This is the HCSM-27 Klein orbit, now derived from ?? 1 alone.*

11 Selection of $L = 8$ from the primitive

Sections 2–10 hold for every even L . The specific value $L = 8$ follows from ?? 1 via the entropy spacing of the torus Laplacian.

Theorem 11.1 (Entropy spacing selects $L = 8$). *Among all integers $4 \leq L \leq 10^4$, the only L for which the $L \times L$ torus has exactly three equally spaced Shannon entropy classes is $L = 8$. At $L = 8$ the three classes are 6.0 bits, 5.5 bits, and 5.0 bits, with uniform spacing 0.5 bits.*

Proof sketch. Compute the Shannon entropy $S(n_1, n_2) = -\sum_x |\phi_{n_1, n_2}(x)|^2 \log_2 |\phi_{n_1, n_2}(x)|^2$ of each mode. For each even L , count the number of distinct entropy classes and their spacing. A direct enumeration over $L \in [4, 10^4]$ gives the stated uniqueness. The full computation is reproduced in [2] and verified at machine precision. $\square$

Remark 11.2 (Independent confirmation). An independent selection of $L = 8$ via maximal winding-generator cancellation is given in HCSM-14b [8]. The two selections use different aspects of the substrate (entropy spacing uses the Laplacian spectrum; maximal cancellation uses the winding generator) and agree at $L = 8$.

12 Physical reading

The theorems of Section 2–11 establish the following chain:

P_0 lattice primitive	(?? 1)
$\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$	(Theorem 11.1)
5-point Laplacian Δ	(Theorem 2.1)
Mirror symmetry $\lambda \leftrightarrow 8 - \lambda$	(Theorem 3.4)
50 paired, 14 residual modes	(Corollary 5.2)
Residual ratio 7/32	(Theorem 7.1)

The physical interpretation of this chain is:

1. The 14 residual modes at $\lambda = 4$ are identified with one Standard Model generation. This is a theorem of HCSM-23 Rev. 7 [9].
2. The residual vacuum energy of the 14 modes is derived in HCSM-53 [12] as

$$\rho_{\text{DE}} = \frac{|F|}{|B|} \cdot m_\nu^4 = \frac{d+1}{2} \cdot m_\nu^4, \quad (31)$$

where $|F| = 10$ and $|B| = 4$ are the fermion and boson sectors of the processor $T = I - P_C$ [7], and $m_\nu = 1.7749$ meV is the neutrino mass from the mass map [10].

3. The numerical value $\rho_{\text{DE}} = 2.4809 \times 10^{-47} \text{ GeV}^4$ matches the observed dark energy density to 0.76%. With the EFT correction $c = 5/9$, $\delta = (1 - \gamma)^2/(2\pi)$ [11, 13], the refined value is $2.5003 \times 10^{-47} \text{ GeV}^4$, matching observation to 0.010% [14, 15].

Remark 12.1 (Scope). This paper does not rederive the mass map, the neutrino mass, the EFT correction, or the promotion mechanism. Those are the subjects of HCSM-33, 52, 53, 54, 56, and 56a. The present paper establishes the substrate, the mirror cancellation, and the residual ratio 7/32—the mathematical foundation on which the physical reading rests.

13 Inputs and derived

13.1 The single axiom

Axiom	Statement
?? 1	Existence of a finite periodic square lattice $\Lambda_L = \mathbb{Z}_L \times \mathbb{Z}_L$ with L even.

13.2 Derived in this paper

Result	Reference
5-point Laplacian uniqueness	Theorem 2.1
Mirror symmetry $\lambda \leftrightarrow 8 - \lambda$	Theorem 3.4
Self-paired eigenvalue $\lambda = 4$	Theorem 4.2
Multiplicity $2(L - 1)$	Theorem 4.3
$50 + 14$ decomposition at $L = 8$	Corollary 5.2
Cancellation theorem	Theorem 6.1
Residual ratio $7/32$	Theorem 7.1
Convention independence	Theorem 8.1
$TF = T_{\text{half}} \circ F$ involution	Lemma 9.3
TF grading by $ \Delta n $	Theorem 9.4
TF-fixed $\Leftrightarrow$ higher $N/4$	Corollary 9.5
Klein four-orbit on $\{u, s, \text{dark}, \tau\}$	Theorem 10.2
$L = 8$ selection	Theorem 11.1

13.3 Imported as theorems of other papers

Result	Source
$14 \leftrightarrow$ SM generation	HCSM-23 Rev. 7 [9]
Processor T and F/B split	HCSM-13 [7]
Mass map and neutrino mass	HCSM-33 [10]
Dark energy density	HCSM-53 [12]
EFT coefficient $c = 5/9$	HCSM-52 [11]
Vacuum stability and γ	HCSM-54 [13]
Vacuum EFT and promotion	HCSM-56, 56a [14, 15]

13.4 Not derived (out of scope)

Item	Reason
The lattice primitive itself	?? 1 is the single axiom
Dimensionful anchor M_P	Input from HCSM-00 [1]
Continuum limit	HCSM-60 (in progress)

14 Reproducibility

All numerical claims are verified in IEEE-754 double precision and are exact in rational arithmetic. The companion verification script `HCSM-10_verification.py` constructs the 64-mode Laplacian, the 14-mode multiplet, the mirror map T_{half} , the diagonal flip F , the TF operator, and the Klein four-orbit, and verifies:

1. Theorem 2.1: the 5-point Laplacian properties.

2. Theorem 3.4: mirror symmetry $\lambda \leftrightarrow 8 - \lambda$.
3. Theorems 4.2 and 4.3: self-paired eigenvalue and multiplicity.
4. Corollary 5.2: the $50 + 14$ decomposition.
5. Theorem 6.1: the cancellation for three test functions.
6. Theorem 7.1: the residual ratio $28/128 = 7/32$.
7. Theorem 8.1: convention independence for affine F with $F(4) \neq 0$.
8. Lemma 9.3: $\text{TF}^2 = \text{id}$.
9. Theorem 9.4: TF-fixed modes are exactly $|\Delta n| = 4$.
10. Corollary 9.5: TF-fixed $\Leftrightarrow$ higher $N/4$.
11. Theorem 10.2: Klein four-group commutation.
12. Theorem 11.1: entropy spacing selects $L = 8$.

Running `12-VERIFICATION/run_all.py` reproduces every numerical claim in the HCSM paper stack.

15 Conclusion

We have derived the HCSM substrate from a single axiom—the existence of a finite periodic square lattice (?? 1). From this primitive:

- The 5-point Laplacian is forced (Theorem 2.1).
- The Laplacian spectrum is exactly mirror-symmetric under $\lambda \leftrightarrow 8 - \lambda$ (Theorem 3.4).
- The unique self-paired eigenvalue is $\lambda = 4$ with multiplicity $2(L - 1)$ (Theorems 4.2 and 4.3).
- At $L = 8$, the 64 modes decompose as $50 + 14$ (Corollary 5.2).
- The residual ratio is exactly $7/32$ (Theorem 7.1), convention-independent for any affine F with $F(4) \neq 0$ (Theorem 8.1).
- The composite involution $\text{TF} = T_{\text{half}} \circ F$ grades the multiplet by $|\Delta n|$ (Theorem 9.4), and TF-fixed status coincides with the HCSM-23 within-pair ordering rule (Corollary 9.5).
- The four modes $\{u, s, \text{dark}, \tau\}$ carry a faithful Klein four-group action (Theorem 10.2).
- The specific value $L = 8$ is selected by entropy spacing (Theorem 11.1).

The physical interpretation—the identification of the 14 residual modes with one Standard Model generation, and of the residual $7/32$ with the observed dark-energy density—is derived in the companion paper HCSM-53 [12], using the mass map of HCSM-33 [10] and the EFT correction of HCSM-52, 54, 56, 56a [11, 13, 14, 15].

Physics is what the machine cannot cancel. The Hodge complex is the machine. The $7/32$ is what the machine cannot cancel.

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